REVISION 2.1 | DATABASE DESIGN SUPPLEMENT

TechConnect Software Requirements Specification

Relational database model for the employed-workforce, company-pricebook and customer-payment operating model

Revision note. This edition retains the supplied Revision 2.0 SRS as its baseline and adds Section 11: a PostgreSQL relational design, entity descriptions, cardinalities, integrity rules and security requirements. Where this supplement details implementation choices, it does not replace the business policies that the SRS leaves open.

Document control

Attribute	Value
Product	TechConnect managed technical services
Baseline source	TechConnect SRS Cameroon, Revision 2.0, dated 22 October 2026
This supplement	Revision 2.1 database design; the preceding baseline SRS is reproduced unchanged
Database target	PostgreSQL 15+; UUID keys; XAF whole-unit amounts stored as BIGINT
Intended readers	Product owner, engineering, operations, price management, payroll and security reviewers

Purpose and boundary

The database is designed around a managed service company: TechConnect employs and dispatches technicians, centrally approves service prices, collects payment, records the work, and pays regular salaries through payroll. Customers request a service and confirm completion. The database supports auditable operations; it does not itself determine applicable employment, tax, privacy or CNPS law. Those calculations and policies require local professional review.

Section 11, appended after the reproduced baseline SRS, groups the database tables by business domain and describes every table in the companion PostgreSQL creation script.

TECHCONNECT

Trustworthy technical services, with employed technicians and transparent fixed prices

SOFTWARE REQUIREMENTS SPECIFICATION | REVISION 2.0

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Revised 22 October 2026 | Cameroon | Initial rollout proposed for selected zones in Douala and Yaounde

Tech-Connect

Product direction

TechConnect is a managed technical-services company, not an open directory of independent tradespeople. Customers request services through the website; employed technicians receive dispatched sessions, diagnose with an approved fault catalogue, and complete work only after the customer approves the fixed-price estimate and payment is confirmed. TechConnect pays technicians through regular payroll and earns a margin on completed services.

Price policy: every selectable service/fault item has a company-approved XAF price. The technician records findings and quantities, but cannot type or negotiate a service price.

Important benchmark distinction: Cameroon's Ministry of Commerce says its Mercuriale is a reference used for public spending and transactions with the State. It is not a blanket nationwide household repair tariff. TechConnect should call its catalogue a standardized company pricebook, using official references where a service/material is covered and transparent market/cost evidence for the rest. [1]

1. Purpose, goals, and scope

This revision supersedes the prior marketplace assumptions. It adopts the owner's employer-led operating model and end-to-end customer journey. It retains the prior SRS's core aims: trustworthy technical work, clear evidence, job tracking, customer recourse, and low-bandwidth access. The brief and previous SRS are context; this document specifies the latest user-authored requirements.

Business goals

- Give customers one accountable service company for technical repair, installation, maintenance, and support.
- Employ and train technicians; pay regular wages independent of each individual job; provide safe work, tools, transport rules, and a path for development.
- Use an approved, consistent pricebook so customers see the same service prices and technicians cannot change them.
- Collect customer payment through the app, deliver the service, pay operating costs and payroll, and retain a sustainable company margin.

Launch scope

The product is for technical home and business services: electrical, plumbing, appliance repair, HVAC/refrigeration, computer/device/network support, and other approved technical trades. Start with a smaller set of high-demand categories in a few zones; expand the catalogue and geography after staffing, quality, and unit economics are proven. "Any service" is a long-term service taxonomy objective, not a day-one promise of coverage.

Out of scope for the first release

Independent provider listings, technician-set quotes, cash-only in-app bypass, platform-held escrow, guaranteed 24/7 response, automatic diagnosis, unapproved work items/prices, and payment to technicians per task in place of salary.

Success measures

MEASURE	PILOT DEFINITION
Reliability	Assigned technician acceptance, arrival and completion rates; report by trade and zone.
Customer confidence	Price shown before repair starts; customer confirmation and test completion recorded.
Workforce stability	Monthly payroll paid on time; staff retention, training completion, safety incidents.
Commercial health	Contribution margin/job, utilization, repeat booking, refunds/rework, and break-even volume.

2. Employment model and responsibilities

Technicians are employees of TechConnect. TechConnect recruits, verifies qualifications, contracts, trains, schedules, supervises, equips, and pays them. Customer funds are paid to TechConnect as the service provider; technicians do not invoice customers or negotiate wages/job prices in the customer session.

Workforce controls

- Create employee records with staff ID, contract status, trade qualifications, skill grades, allowed job types, service zones, shifts, leave, equipment issue/return, training, and safety incidents. Restrict HR and identity fields from customer-visible profiles.
- Assign work based on required skill/authorization, service zone, availability, workload, and fair rotation; record dispatch reason and assignment history.
- Provide fixed monthly salary according to a locally reviewed pay band, applicable minimum wage/collective terms, hours and overtime rules, and benefits. A quality/safety bonus may supplement salary after HR/legal review; it must not turn a timer into piece-rate pay.
- Maintain payroll runs, salary statements, attendance/approved leave, overtime approval, allowances, deductions, and payslip access. Job timers support service tracking, not wage calculation.
- Support worker safety: supervisor escalation, check-in/arrival, worksite hazard notes, protective equipment, incident reporting, emergency contact, and travel policy.

Cameroon compliance note

The CNPS states employers must register workers within eight days of hiring, report hiring/termination, declare salaries monthly by the 15th of the following month, and pay social contributions within the prescribed post-month deadline. The product should produce reliable payroll/CNPS records, but a payroll officer and local counsel must validate calculations, contracts, wage floor, collective agreements, taxes, and reporting rules before production. [2]

The Prime Minister’s 2024 decree sets a SMIG for State employees governed by the Labour Code; do not copy that state-specific figure as the private-company wage schedule. TechConnect’s applicable wage band must be validated for the company’s sector and technician grades. [3]

Employee workspace

Technician dashboard shows assigned sessions, address only for authorized assignments, customer problem notes, route/contact actions, diagnosis workflow, accepted work, job timer, approved pause controls, completion submission, and own schedule/training. It does not show customer payment card/mobile-money secrets or allow price edits.

3. Standard service pricebook

TechConnect owns and versions the pricebook. A centrally approved catalog contains every billable diagnosis, repair, installation, maintenance task, labor unit, and approved parts/material item. Technicians select codes and quantities; the system calculates the total.

National reference prices versus company price

The Ministry of Commerce describes the Mercuriale as a public expenditure control/reference catalogue for transactions with the State. It can inform eligible items used in public contracts, but it does not establish a universal consumer retail price for every household repair. TechConnect can standardize its own customer prices nationwide, but should not label them government-mandated national tariffs unless a specific applicable regulation says so. For all other items, construct company rates from observed local cost, labor time, materials, transport, tax, warranty reserve, and margin; disclose them as TechConnect prices. [1]

Pricebook record requirements

FIELD	RULE
Service/fault code	Unique stable code; category/trade; bilingual short and detailed description; fault symptom and included scope.
Price and unit	XAF amount, unit (fixed / per hour / per meter / per item), minimum billable quantity, tax treatment, effective dates.
Coverage	Travel/diagnostic fee rule, included labor, approved parts/consumables, exclusions, warranty and customer preparation.
Governance	Draft, reviewer, approval, source/benchmark, version, approval timestamp and change note. No technician editing.
Customer display	Item, quantity, unit price, subtotal, discount if policy permits, tax, total payable and validity window.

Price calculation

Approved estimate total = call-out/diagnostic fee (if disclosed and applicable) + sum(approved catalog item price x approved quantity) + approved parts/materials + disclosed tax/fees - authorized discount. Only catalog items or centrally approved exceptions can become billable. An unlisted item is blocked from work/payment until pricebook administration creates or approves an item and the customer accepts a new estimate.

Pricebook administration requirements

- Only authorized price managers can draft/publish catalog versions; second-person approval for price changes.
- Publish effective date and preserve the exact version on every customer estimate/invoice.
- Show customers the selected service’s price or fee rules before they submit a request; show the complete itemized repair total before taking payment.
- Record source benchmark, cost assumptions, tax treatment, review date, margin floor and exception reason per price item.
- If a price change occurs after diagnosis, never silently change an open estimate; reissue it and require fresh customer consent.

4. Customer and technician workflow

The workflow below follows the owner's requested order while separating the initial dispatch/diagnostic fee from the final repair estimate. Product owner must approve the cancellation/refund policy and any visit fee before launch.

STATE	ACTOR / SYSTEM ACTION	GATE / RECORD
1. Browse	Customer logs in, browses service categories, chooses the service and may select an available/qualified technician.	Availability, service area, indicative fixed fees and expected arrival are visible.
2. Request	Customer enters name/contact, problem description and full service address, selects slot and creates session.	Required fields validated; consent/privacy notice; unique session ID; address restricted to dispatch.
3. Dispatch	Session appears in eligible technicians' dashboard; selected technician accepts or declines. System may reassign.	Assignment, response time and reason logged; client sees confirmed technician and ETA.
4. Travel/arrival	Technician reviews authorized issue/address, prepares, starts travel, arrives and checks in.	Status/time events recorded; customer notified. Travel time is not repair timer.
5. Diagnose	Technician inspects fault and selects approved symptoms/fault codes, tasks, quantities and parts from checklist/catalog.	No free-text prices; photos/notes optional; technician attests diagnosis. Unlisted item routes to supervisor.
6. Estimate	System calculates fixed price lines and total, creates immutable itemized estimate; technician submits to customer.	Pricebook version, codes, quantities, diagnosis, warranty, exclusions and validity frozen.
7. Consent/payment	Customer reviews diagnosis/line items, accepts or declines. If accepted, pays the displayed amount in app.	Record explicit acceptance. Work authorization only after payment success callback and receipt.
8. Repair timer	Technician taps Start Work after payment confirmed; estimated duration timer begins.	Work start timestamp; expected duration from approved task catalog. Pause requires reason and customer notice.
9. Technician finish	Technician records tasks/parts, evidence and outcome, taps Work Complete.	Stop elapsed timer; customer testing starts; status is awaiting customer confirmation.
10. Customer test	Customer tests the service and presses Complete, or reports a defect/rework/dispute.	Only customer completion closes session; if disagreement, rework/dispute state remains open.

5. Functional requirements

MUST requirements are launch gates; SHOULD items are planned pilot capabilities. Each requirement must map to an acceptance test.

ID	REQUIREMENT
FR-01 MUST	Customer can browse approved service categories and see fee/price rules, coverage, and availability before requesting.
FR-02 MUST	Request requires customer name/contact, issue description, full service address, service selection, and preferred time; consent and session ID stored.
FR-03 MUST	Session is dispatched only to employees authorized by trade, skill level, zone, schedule, and workload; staff can accept/decline with reason.
FR-04 MUST	Assigned technician can view customer address/problem for active assignment only; reassignment revokes access when no longer needed.
FR-05 MUST	Technician records diagnosis by approved fault/task checklist and quantity; no price entry, custom billable line, or unapproved task.
FR-06 MUST	System calculates estimate only from active catalog version; freezes the itemized snapshot and sends it to the customer.
FR-07 MUST	Customer must explicitly accept the diagnosis/estimate before payment; declined estimate cannot enter repair work.
FR-08 MUST	Payment uses an authorized payment provider; successful server-verified payment unlocks work start; duplicate callbacks are idempotent.
FR-09 MUST	Repair timer starts only after payment success and technician Start Work; planned duration and elapsed time are separately stored.
FR-10 MUST	Pause/resume records reason, actor, time, customer notice and supervisor escalation; approved waits do not count against speed metric.
FR-11 MUST	Technician completion submits work summary, catalog lines, parts, evidence, time and outcome; status waits for customer test.
FR-12 MUST	Customer can confirm completion, request included rework, or open dispute; only customer confirmation closes session.
FR-13 MUST	Generate receipt/invoice with company details, catalog version, line items, tax, total, payment status and warranty terms.
FR-14 MUST	Rating is requested after customer completion; tie feedback to job; allow review moderation and appeal.
FR-15 MUST	Staff can create employee profile, role, trade authorization, shift, zone, training and active/inactive status.
FR-16 MUST	Payroll module/export records salary, allowances, attendance/overtime approvals, deductions, payslip, monthly declaration and CNPS export fields.
FR-17 MUST	Central admins version price catalog; maker-checker approval; effective dates; audits; immutable estimates.
FR-18 MUST	Refund/cancel/rework workflows enforce policy and preserve provider callback IDs, decisions and audit.
FR-19 SHOULD	Customer dashboard shows assigned technician, ETA, session stages, estimate/payment receipt, timer status and history.
FR-20 SHOULD	Supervisor console supports dispatch queue, reassignment, unlisted-catalog approval, late-job alert, complaint and safety escalation.
FR-21 SHOULD	Monthly technician dashboard shows schedule, completed jobs, quality metrics, training, and salary statement without tying salary to a job timer.

6. Payment, timer, rating, and completion rules

Payment is charged to TechConnect for the accepted estimate through an approved provider integration. Store provider token/reference and status, never raw card credentials or Mobile Money PIN/OTP. Work cannot start on a client-side “success” screen alone; the backend must validate signed provider confirmation.

Payment states

CREATED -> PENDING -> SUCCEEDED, FAILED, EXPIRED, REFUND_PENDING, PARTIALLY_REFUNDED, REFUNDED, or DISPUTED. Bind each payment to one session and estimate version. Use idempotency keys, signed webhooks, reconciliation, receipt creation, duplicate detection, and daily settlement checks. If payment fails, keep job in awaiting payment; do not release Start Work.

Estimate changes and extra work

If diagnosis changes or more faults are found after an accepted estimate, technician pauses, selects additional approved catalog items, and submits a revised version. Customer must approve the revised total and pay any difference before related extra work starts. Decreases trigger a partial refund/credit under the approved policy. No verbal-only extra work.

Timer semantics

- Planned minutes come from the selected catalog line items and configured complexity bands. Actual elapsed work time begins only when payment is verified and technician taps Start Work.
- Timer is a service-level measurement, not a salary meter. Salary remains regular payroll.
- Pause requires a reason code (customer unavailable, safety stop, missing approved part, scope change, site condition, other); notify customer and preserve both active-work and wall-clock durations.
- Speed score adjusts for approved pauses and job complexity; do not reward unsafe rushing or penalize valid diagnostics.

Customer acceptance and guarantee

On Work Complete the client sees a checklist to test the repair, review work summary, and choose Confirm Complete, Request Included Rework, or Report Problem. A configurable acceptance window can send reminders, but session must not auto-close during an active report. Warranty/rework eligibility is linked to catalog item, invoice, parts and completion timestamp.

Rating dimensions

Record customer feedback on technical quality, communication, care/safety, punctuality, and satisfaction. Skill/experience derive mainly from verified qualifications, training and authorized job history; speed derives from adjusted completion data. Show sample size and separate measures. A single negative rating does not automatically reduce salary or terminate employment; human review and employee response/appeal are required.

7. Business model and unit economics

TechConnect sells a managed service: it collects customer revenue, supplies an employed workforce and tools, pays regular salaries/social charges, and keeps the residual margin after service costs. The technician earns salary for employment and may be eligible for a separately governed quality/safety incentive.

Per-job economics

Net service revenue = customer payment excluding taxes and pass-through charges. Contribution per completed job = net service revenue - parts/consumables - payment-provider fees - direct travel/dispatch cost - allocated technician labor time - warranty/rework reserve. Monthly operating profit = total contribution - fixed payroll not already allocated - supervisors, rent, tools depreciation, insurance, software, customer support, marketing, taxes and other overhead. Break-even completed jobs/month = monthly fixed operating costs / average contribution per completed job.

Pricing and payroll guardrails

- Maintain a finance-approved pricebook with cost basis and minimum contribution margin by zone/trade; reprice centrally when supplier, wage, fuel, tax, or payment costs move.
- Budget paid, productive, travel, training, leave, and idle time. Salaried capacity is a fixed cost, so workforce rosters must match demand forecasts and service-level coverage.
- Keep customer payment reconciliation separate from employee payroll approvals; no mechanic receives customer money at the door.
- Measure monthly utilization (billable service time / paid available time), contribution/job, first-time fix, rework reserve, payroll coverage, and cash runway.
- Do not promise profit or fixed monthly wage until service volumes, pay bands, taxes and operating costs are validated with pilot data.

Revenue sources

Primary revenue is the fixed catalog price charged for completed work; optional maintenance plans or business contracts may be introduced later with separate pricing and service-level terms. Any discounts must be authorized centrally and visible on the estimate; technician cannot offer side discounts or collect unrecorded payment.

Pilot plan

Recruit a small salaried team across a few trades and zones. Validate a defined pricebook, dispatch utilization, job completion, first-time fix, customer repeat, payment settlement, payroll burden, and warranty costs before scaling headcount. Start with manual dispatch and reconciliation support, but keep every session/payment/payroll event in the system.

Tech-Connect

8. Data model, architecture, and privacy

Recommended MVP architecture: responsive low-bandwidth PWA; role-based API; modular monolith for catalog, dispatch, sessions, estimate, payments, completion, ratings and payroll export; managed relational database; private media storage; async notification/PSP callback workers; operations/admin console.

Core entities

Customer, Employee, EmployeeTradeAuthorization, Shift, ServiceCategory, CatalogItem, PricebookVersion, ServiceRequest, DispatchAssignment, DiagnosisFinding, EstimateVersion, PaymentAttempt, WorkTimerEvent, SessionEvent, WorkEvidence, CompletionDecision, Rating, ReworkCase, Refund, PayrollPeriod, PayslipReference, CNPSExport, AuditEvent. Each job links to the exact catalog/version and employee authorization used.

Security and data handling

- Encrypt transport and stored data; admin MFA; staff role scopes; audit all customer address access, price changes, payment refunds, payroll edits, and employee evaluation.
- Collect full address because the service requires dispatch, but show it only to assigned staff with an active job. Record access and revoke on reassignment/closure. Customer privacy notice explains purpose, sharing with assigned technician, retention and rights.
- Store no payment secrets. Use provider-hosted tokenized checkout or approved SDK; validate callbacks server-side. Separate payroll/HR data from operational dashboards.
- Strip EXIF from photos; signed media links; retention periods for ID, address, job evidence, payment receipts and payroll records configured to applicable law and tax/accounting needs.
- Cameroon enacted Personal Data Protection Law No. 2024/017; confirm current duties/implementation with local counsel and build notices, access/correction, retention and incident handling. [4]

Non-functional targets

AREA	TARGET
Performance	p95 under 2.5 seconds for browse/session/estimate API at pilot load, excluding external PSP/network latency.
Availability	99.5% monthly for core session API during pilot; alert operations on dispatch/payment outage.
Reliability	Idempotent session, estimate, payment and webhook operations; daily payment reconciliation; tested backups.
Connectivity	Core pages compact; requests saved/resumable; notification fallback does not lose assignments.
Localization	French/English, XAF, Cameroon phone/date/time; clear taxes/fee display.

9. Acceptance tests, rollout, risks

Release acceptance tests

- A customer cannot submit a request missing name/contact, problem description, address, service category, or consent; session appears in the dispatch queue with unique ID.
- Only an authorized active employee can accept the dispatch. Customer sees confirmed assignment; former assignee loses address access after reassignment.
- Technician diagnosis accepts only active catalog fault codes/quantities. Attempts to enter arbitrary service price or add an unapproved code are rejected and audited.
- Estimate lines resolve to catalog version and item prices. Customer accepts explicit total; declined estimate cannot proceed to payment/work.
- Payment success callback verified by backend unlocks Start Work exactly once. Failed/pending/duplicate callback does not start timer twice or authorize repair.
- Timer starts after payment and technician action. Pause/resume records reason and customer notice; payroll output is independent of timer duration.
- Technician Work Complete moves job to customer testing. Only customer Confirm Complete closes; Request Rework/Report Problem routes to open case without losing evidence.
- Receipt, rating invitation, warranty/rework deadline, payment reconciliation and salary/reporting exports are generated from auditable records.

Rollout

Phase 1: field discovery and approved catalog for limited trades/zones; collect supplier/labor/travel costs and validate customer understanding. Phase 2: staff/payroll/dispatch and PWA MVP with manual supervisor support. Phase 3: controlled pilot; measure service quality, job contribution, technician utilization, payroll, and payment/rework. Phase 4: add trades and locations only when workforce capacity and unit economics meet targets.

Key risks

RISK	CONTROL
Customer interprets company rates as government tariffs	Label catalog “TechConnect standard prices”; reference official Mercuriale only within scope; counsel reviews claims.
Underpriced items make salary model unprofitable	Costed pricebook, margin floor, scheduled repricing, zone-level job-costing and pilot gates.
Technician rushes because timer/rating rewards speed	Timer not linked to wage; quality/safety weighted higher; adjust for approved delays; supervisor reviews outliers.
Wrong diagnosis / unlisted fault	Approved checklist, supervisor escalation, new estimate and fresh customer approval before work.
Payroll or CNPS non-compliance	Payroll controls, reminders/exports, accountable HR/payroll owner and local labor counsel review.
Payment provider outage or disputed charge	State machine, reconciliation, clear status, no work authorization until confirmed, refund/support process.

10. Open decisions and sources

Decisions required before production

- Which trades and city zones launch first; what licensing/certification evidence is mandatory by trade?
- Which exact catalog items and units are public-sector Mercuriale matches, and which are TechConnect cost-based rates? Who approves changes and effective dates?
- What are the customer diagnostic/dispatch fee, cancellation, refund, parts procurement, warranty, rework and out-of-catalog exception policies?
- Which employment contract, pay bands, schedules, overtime, leave, safety, CNPS, tax and payroll rules apply to each staff category? Obtain local professional review.
- Which licensed payment provider supports customer collection, refunds, settlement reconciliation and receipts? Will the displayed price include VAT and mandatory charges?
- What customer acceptance timeout/reminders apply, and how are unresponsive clients or suspected damage handled?

Evidence and references (accessed 22 October 2026)

[1] Cameroon Ministry of Commerce, La Mercuriale. Official page states this price reference is for controlling public expenditure and transactions with the State; it should not be represented as a universal household repair tariff.

[2] CNPS, Employer obligations. Worker registration, monthly salary declaration and contribution process.

[3] Prime Minister's Office, Cameroon, Decree No. 2024/0168/PM of 23 Feb 2024 establishing the SMIG applicable to State employees governed by the Labour Code; not a private-company-wide service price or wage table.

[4] Presidency of Cameroon, Law No. 2024/017 on Personal Data Protection, 23 Dec 2024.

[5] Cameroon Ministry of Commerce, Law No. 2010/021 on electronic commerce.

[6] Revised source document: TechConnect SRS Cameroon, edited edition supplied by the user. This revision changes the independent marketplace/direct-pay design to an employed-workforce/company-pricebook/app-payment model.

Revision note

This revision replaces independent providers, technician-set quotes and direct-to-technician payment with employee dispatch, controlled diagnosis checklists, fixed company catalog pricing, customer in-app payment, a post-payment work timer, regular salary payroll, and customer-confirmed completion. The pricebook is a company policy grounded in applicable benchmarks; the official Mercuriale has a stated public-sector scope.

11. DATABASE DESIGN

11.1 Relational model overview

The model separates identity and workforce data from customer operations; versioned catalogue data from estimates; payment evidence from work authorization; and service timing from salary. A service request is the durable job/session record. Events and assignment records preserve how that job changed over time.

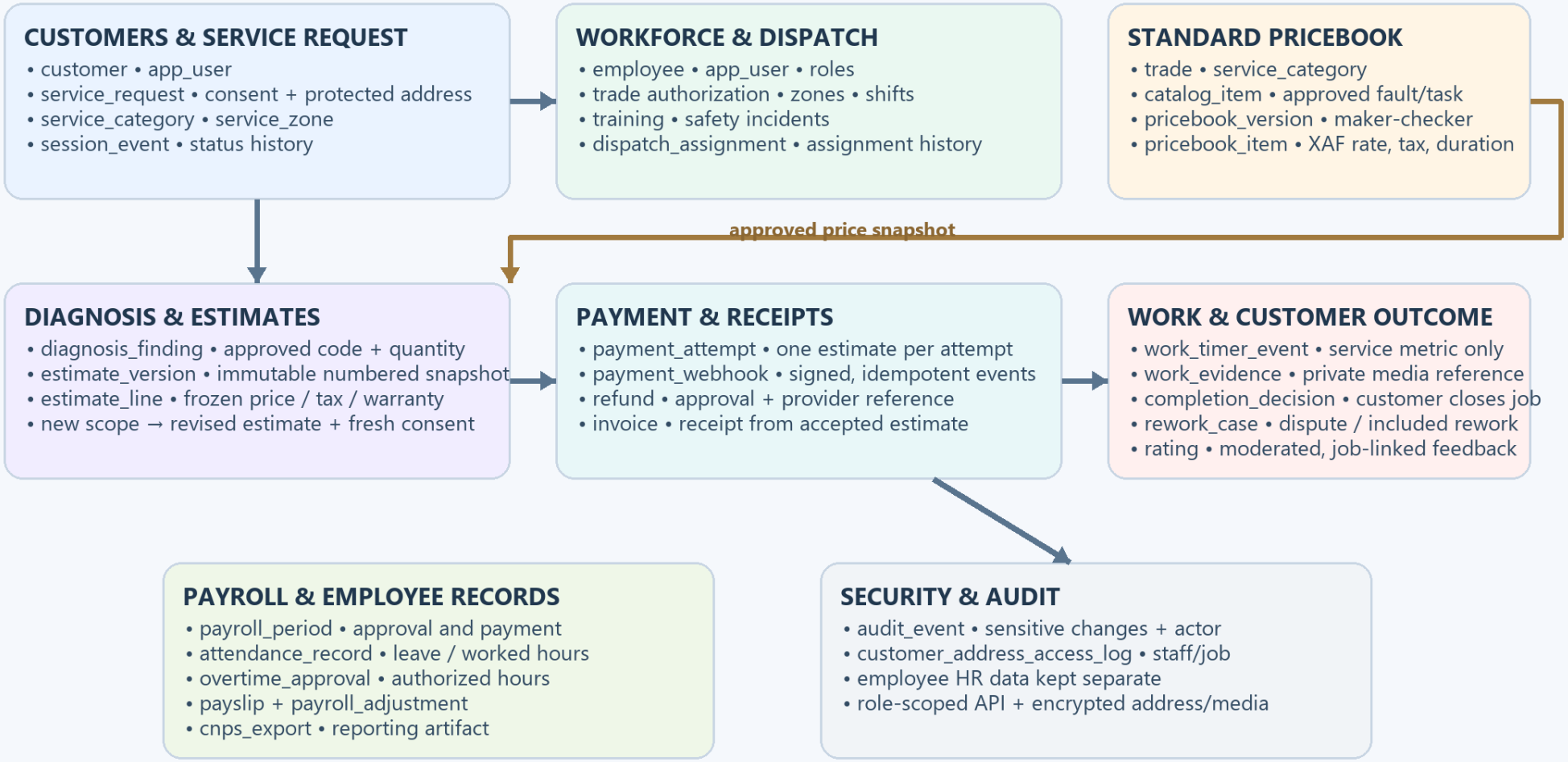
Figure 11-1 is shown on the following landscape page to make the major database groups and relationships legible. The entity catalog and detailed rules follow.

Relationship cardinalities

Relationship	Cardinality / meaning
Customer → service request	One customer may submit many requests; each request belongs to one customer.
Request → assignments / findings / estimates	One request can have many historical assignments, diagnosis findings and numbered estimate versions.
Pricebook version → pricebook item	A version contains priced catalogue entries; an item code can have different approved prices across versions.
Estimate → estimate lines → payment attempts	An estimate freezes one or more item lines; payment attempts reference the exact estimate being charged.
Request → work evidence / decisions / cases	Operational evidence and customer decisions are retained against the request; a report or rework case does not close it.
Employee → authorization / payroll	An employee can hold multiple dated trade authorizations and receives payslips for payroll periods.

TECHCONNECT | DATABASE MODEL

PostgreSQL relational design • Core entities grouped by business domain



CONTROL POINTS

Price is centrally approved • Work starts only after verified payment • Customer confirmation closes the job

Salaries are processed through payroll and are independent of job timer duration.

11.2 Entity catalog — identity, workforce and pricebook

Storage conventions. Business records use UUID primary keys unless a short role key is appropriate. Money is stored as integer XAF francs; quantities and fractional tax rates use NUMERIC. Timestamps use TIMESTAMPTZ and display in the customer/staff locale. Addresses are ciphertext and are decrypted only after active-assignment checks.

Identity and access

Entity	Description and important attributes
app_user	Authentication/contact identity shared by customers and staff. Unique email/phone, preferred language, account status and creation time. Stores password hash only if local authentication is used; otherwise federate authentication.
customer	Customer profile keyed to app_user; display name and creation time. Request-level snapshots preserve the contact details used at booking.
employee	Staff record keyed to app_user; unique staff number, legal name, employment status, hire/termination dates, contract type, emergency contact and restricted HR data. Keep HR fields out of operational views.
app_role / user_role	Role catalogue and many-to-many grants for customer, technician, dispatcher, supervisor, price manager/approver, payroll officer and administrator. Application authorization must enforce least privilege.

Workforce qualification and scheduling

Entity	Description and important attributes
trade	Approved trade taxonomy (e.g., electrical or plumbing) with stable code and French/English names.
employee_trade_authorization	Dated employee authorization for a trade, skill grade, qualification evidence, approver and approval timestamp. Dispatch must validate that authorization is active for the job.
service_zone / employee_zone	Launch coverage areas and dated employee eligibility for each zone. Service requests reference one zone; assignment checks current coverage.
employee_shift	Scheduled employee interval and state (scheduled/completed/cancelled/leave). Used with workload and availability during dispatch.
employee_training	Training course, provider, completion/expiry dates and evidence reference. Supports employee development and qualification review.
safety_incident	Employee safety report, optional related request, reporter, type, description, severity and investigation state. It is restricted HR/safety data.

Company pricebook and approved catalogue

Entity	Description and important attributes
service_category	Customer-facing service grouping tied to a trade. Bilingual names and stable category code.
catalog_item	Approved fault, labor task, part, material or fee with stable code, bilingual description/symptom, included scope and active state. Technicians select codes; they cannot add a price.
pricebook_version	Governed catalogue release with unique version code, status, effective dates, creator, reviewer, approver, approval timestamp and change note. Maker-checker approval requires a different approver from the creator.
pricebook_item	Version-specific price and unit for a catalog item: XAF unit price, minimum quantity, tax code/rate, planned minutes, warranty days, exclusions, source benchmark, cost assumptions, margin floor and review date.

Section 11.3 — Request, diagnosis and estimate

Service request and dispatch

Entity	Description and important attributes
service_request	The durable customer job/session. Stores public reference, customer/category/zone, requested service and slot, problem description, contact snapshot, encrypted address, consent version/time, lifecycle status and timestamps. The address is only exposed to eligible active assignees.
dispatch_assignment	Append-only assignment history: request, employee, qualification used, dispatcher, reason, offer/response/revocation timestamps, acceptance/decline/reassignment status and ETA. Former assignees lose address access when no longer active.
diagnosis_finding	Technician finding against an approved catalog code and assignment, with observed symptom, quantity, notes and technician attestation. Supervisor approval fields support unlisted-item escalation; no free-text price is stored.
session_event	Append-only lifecycle transition with actor, previous/next status, reason, metadata and occurrence time. Supports customer tracking and audit reconstruction.

Estimate snapshots and price integrity

Entity	Description and important attributes
estimate_version	Numbered estimate for a request, tied to one pricebook version. Stores status, fees, discount, tax, total, XAF currency, warranty/exclusions, validity, submission and customer decision, plus creator. Submitted or accepted estimates are immutable.
estimate_line	Frozen line-level evidence: referenced pricebook/catalog item, code and bilingual description snapshot, quantity, unit, unit price, tax rate, subtotal, planned duration and warranty. Snapshot remains interpretable after future pricebook changes.

Request-to-estimate sequence

A request is validated with consent and address before entering dispatch. An eligible employee accepts the assignment and records approved diagnosis codes and quantities. The server calculates an estimate using one effective, published pricebook version and copies all customer-facing values into estimate lines. A customer decision is explicit. Changed scope creates a higher numbered estimate version and requires fresh consent; payment of any added amount is recorded before related work starts. Decreases follow the centrally approved credit/refund policy.

Required integrity checks

The API/database transition layer must verify that every estimate line belongs to the estimate’s pricebook version, its quantity meets the item minimum, the version is effective and published, and the displayed total equals call-out fee + line subtotals + disclosed tax/fees − authorized discount. These cross-row calculations need a transaction-safe service or stored procedure; ordinary row CHECK constraints cannot prove them.

Section 11.4 — Payment, work and customer outcome

Entity	Description and important attributes
payment_attempt	One provider collection attempt bound to a request and exact estimate. Stores provider, unique idempotency key, provider reference/token reference, XAF amount, status and server verification time. Never store raw card data, Mobile Money PIN or OTP.
payment_webhook	Provider event ID, payment link, signature validation result, payload, receive/process timestamps and result. Unique provider + event ID makes duplicate callback processing idempotent. Retain payload securely under defined retention.
refund	Payment-linked amount, reason, requester, approver, provider reference, state and resolution timestamps. Supports approved cancellation, partial-refund and dispute paths.
invoice	Unique invoice number, request and estimate references, company details/totals snapshots, pricebook version code, warranty terms and issue time. Receipt/invoice remains consistent with the charge that was accepted.
work_timer_event	Start/pause/resume/stop event linked to request and assignment. Captures actor/time, pause reason, customer notice, supervisor escalation and payment proof reference. This is a service metric and is not a payroll input.
work_evidence	Private media storage key, uploader, media type, digest, caption and EXIF-stripped flag. Store media outside the relational database behind short-lived signed links.
completion_decision	Customer’s confirm-complete, request-rework or report-problem choice with timestamp and notes. Only confirm-complete closes the service request.
rework_case	Request, original estimate, opener, reason, open/resolved/disputed state and resolution. Preserves evidence and links rework to the original warranty/estimate.
rating	One feedback record per request, tied to customer and employee, with five independent 1–5 dimensions, comment, moderation state/note and timestamp. Rating alone must not automatically change pay or employment.

Payment and work authorization rules

Payment transitions follow the SRS state machine: created → pending → succeeded, failed, expired, refund_pending, partially_refunded, refunded or disputed. Only a valid server-verified success callback for the accepted estimate unlocks Start Work, exactly once. Start time also requires the technician’s action. Failed, pending or duplicate callbacks do not authorize work or start a timer. A revised estimate must be accepted and any difference settled before extra work.

Work completion remains pending customer testing after the employee submits the work summary. Customer confirmation closes the request; a defect, rework request or dispute leaves it open. Warranty eligibility is determined from the estimate/invoice snapshot, catalog warranty, parts and completion time.

Section 11.5 — Payroll, audit and reporting

Entity	Description and important attributes
payroll_period	Start/end dates and draft/review/approved/paid/declared state. Records payroll approval, payment date and CNPS declaration timestamp.
attendance_record	Employee/date/period, worked/leave/sick/holiday/absence/training category, hours and approval status/approver.
overtime_approval	Employee and payroll period, approved overtime hours, reason, approver and approval timestamp. Payroll must use approved records, not job timer duration.
payslip	Employee-period statement with gross, allowances, deductions, net, employee/employer CNPS and withholding totals in XAF, plus private payslip storage reference. One per employee per period.
payroll_adjustment	Separately approved allowance/deduction/bonus/correction line for a payslip, with amount, reason, approver and timestamp.
cnps_export	Generated CNPS-period export reference, creator, generation/submission time and submission reference. Schema records/export fields; professional review validates the calculation and filing rules.
audit_event	Actor/action/entity, reason, before/after snapshots, timestamp, request correlation ID and IP. Required for price changes, address access, payroll edits, refunds and employee evaluation. Append-only access is recommended.
customer_address_access_log	Employee, request, assignment, timestamp and reason for address access. Access service first verifies assignment is active; reassignment/closure revokes permission.
app_role / user_role	See 11.2. These grants are a control input and should be synchronized with API scopes and database roles.

Operational reporting supported

The model can produce assignment acceptance/arrival/completion rates by trade and zone; time to respond and arrive; first-time-fix, rework, refunds and customer ratings; pricebook version and margin review; payment settlement and duplicate callback reports; paid availability and utilization; payroll paid-on-time and CNPS export tracking. Timer-derived speed measures require approved-pause and complexity adjustments and must remain separate from wage decisions.

Retention and privacy

Encrypt transport and sensitive stored fields; use managed key encryption for addresses and private HR data. Restrict access with role scopes and, where appropriate, PostgreSQL row-level security. Keep identity documents, addresses, evidence, receipts, payroll and audit records under configurable retention schedules validated against applicable Cameroon law. Log administrative reads and sensitive changes; strip photo EXIF; keep media in private object storage.

Section 11.6 — Database requirements and acceptance criteria

ID	Requirement / acceptance condition
DB-01	Referential integrity: a request references an existing customer, category and service zone; assignment and estimate history cannot be orphaned.
DB-02	Catalog control: only authorized, effective, published pricebook items can appear in submitted estimates; price editing is unavailable to technician roles.
DB-03	Snapshot immutability: submitted estimate lines retain price, description, quantity, tax and warranty values after catalog changes. Revised scope creates a new estimate version.
DB-04	Maker-checker: pricebook creator and approver are distinct; approval actor and time are auditable.
DB-05	Payment safety: unique idempotency keys and provider event IDs; invalid signature, duplicate or failed callbacks cannot authorize work.
DB-06	Timer/payroll separation: payroll generation uses approved attendance, salary and adjustments; no timer duration contributes to base salary calculation.
DB-07	Closure rule: only a customer confirm-complete decision closes a request; rework/dispute remains open and retains evidence.
DB-08	Privacy: customer address access requires a current assignment and is logged; reassignment and closure revoke access. HR/payroll data is not available in technician/customer views.
DB-09	Auditability: capture actor and time for price changes, dispatch, estimate consent, payment/refund, work transitions, payroll edits and moderation.
DB-10	Recovery: backups are encrypted and restoration is periodically verified; session/payment/webhook operations are idempotent and reconciliation is available.

SRS traceability

The schema directly supports FR-02 to FR-18: required request and consent; qualified dispatch; restricted address; approved diagnosis; versioned estimates; explicit consent; server-verified payments; timer and pause records; completion evidence; customer decision; invoice; rating/moderation; employee authorization; payroll/CNPS export; maker-checker price governance; and cancellation/refund/rework audit. FR-19 to FR-21 can be built from the same records for customer tracking, supervisor console and employee/salary dashboards.

Open product decisions remain open

Before production, approve launch trades/zones and qualification evidence; item prices, taxes, fees and pricebook authority; cancellations, refunds, parts, warranty/rework and acceptance timeout; PSP and settlement procedure; employment contracts, pay bands, overtime/leave, tax and CNPS calculations; and privacy/retention rules. This database design intentionally provides records for these policies without inventing their values.

Implementation note

The companion SQL file is a PostgreSQL 15+ schema foundation, not a fully deployed application. Before production, add migrations, database grants, row-level security, state-transition procedures/triggers, immutable-record protection, encryption/KMS wiring, indexes informed by pilot load and tested backup/restore procedures. The SQL DDL has not been applied to a live PostgreSQL server.